

Group Name: Nova Six

Project Leader	Sarthak Sinha, 9785565, sarthaksinha2k5@gmail.com
Project Members	Cheung Xi Min, 1441164, cxmin0805@gmail.com
	Wang Wenjie, 1441607, leewenjie215@gmail.com
	P Reshmah, 1349077, preshmah15@gmail.com
	Loh Jun Heng, 1366956, exceedlohjunheng@gmail.com
	Remo Jhaison Jayakumar, 9088477, remorockstar1126@gmail.com

Figma Lo-Fi & Hi-Fi link:

Lofi :

<https://www.figma.com/design/rUbwkHeSEpaxyMIZfwwj1i/CSIT226-Nova-6-Project?node-id=0-1&p=f&t=sw0BvPIDOnfnJnOE-0>

Hifi:

<https://www.figma.com/design/rUbwkHeSEpaxyMIZfwwj1i/CSIT226-Nova-6-Project?node-id=0-1&p=f&t=sw0BvPIDOnfnJnOE-0>

SDG 15 - Life on Land

“Nurturing Future Leaders in Animal Protection”

Modified Part A

(Modified from marker's remark)

Grade item	Calculated weight	Grade	Range	Feedback	Contribution to site total
▼ CSIT226 (SP226) Human Computer Interaction					
QUIZ Asgn1_Quiz_1	-	27.00	0-30	-	-
QUIZ Asgn1_Quiz_2	-	-	0-30	-	-
ASSIGNMENT Asgn 2 Part A Submission	-	80.00	0-100	-	-
ASSIGNMENT Asgn 2 Part B Submission	-	-	0-100	-	-
ASSIGNMENT Asgn 2 Part C Submission	-	-	0-100	-	-

Feedback not found

App Functionalities

Main Page	- The user portfolio is located on the top left of the screen, with the user's profile picture displayed in a small circle. When a user clicks into this, it goes to their profile page which is a summary of what the user has achieved and topics covered. - Map showing locations of an animal once it has been virtually adopted through the donation page. - Navigation bar, makes navigation between different pages seamless. - A small square which shows how many days the user had logged on consecutively. - Right beside the summary menu is a “continue from where you stopped” bar, which displays the topic which the user last visited in their previous logging. When the user taps on this function, it will jump to the page which was last accessed (only for learning topics) - A progress bar which displays in percentage, how many topics the user had covered and completed. - An achievement bar which displays the latest 3 achievement badges the user had earned through learning and quizzes. - A “Take a quiz” tab allows the user to jump to the quiz which they last attempted.
Navigation bar	- Contains the following options: 1) Home 2) Learn 3) Donate to a cause

	4) Settings
Learn About Biodiversity	- A page with multiple soft rectangular buttons, containing a picture of a specific habitat - Each habitat will cover on the weather, plants and animals which lives in it - Once the user completes each topic, there will be a short 5 question MCQ pop-quiz to help users refresh on the habitat covered.
Quizzes	- After the user has completed the topic, they will be brought to a summary page which summarises what the topic has taught so far. - At the bottom of the summary page will have a small pop up which directs the user to the quiz relating to the topic they just learnt. - To keep the app engaging without the classroom feel, the quizzes are designed to be short, but covering all the important details. - Users are able to compete their quiz scores with friends, integrated scoreboards displays who scored higher in each quiz.
Donate to a cause	- This page allows users to choose an amount as well as a method to donate to conservation efforts in Indonesia. - A message tab will be displayed at the top, notifying users that all donations made will be used towards conservation efforts. - An option for user to directly input their bank card detail will be displayed near the top of this page. This decision allows users to make donations easily. - If users feel uneasy saving their bank details in the app, other common methods are also implemented into the app. These options are: Apple Pay, Samsung Pay, Google Pay, Paynow and Paylah. - Users are allowed to choose whether they would like to donate every month, or only make a one-time donation. - This page will allow users to adopt a virtual animal after a donation is made. Once an animal has been adopted, an icon of the animal will be displayed on the map in the homepage.
Settings	- Users are able to customize the app to a certain extent, as well as manage their privacy settings.

Overall Design Considerations

1. The color scheme will be using different shades of green and white.
2. To make the interface more attractive to users, taking inspiration from the IOS UI design of soft shapes.

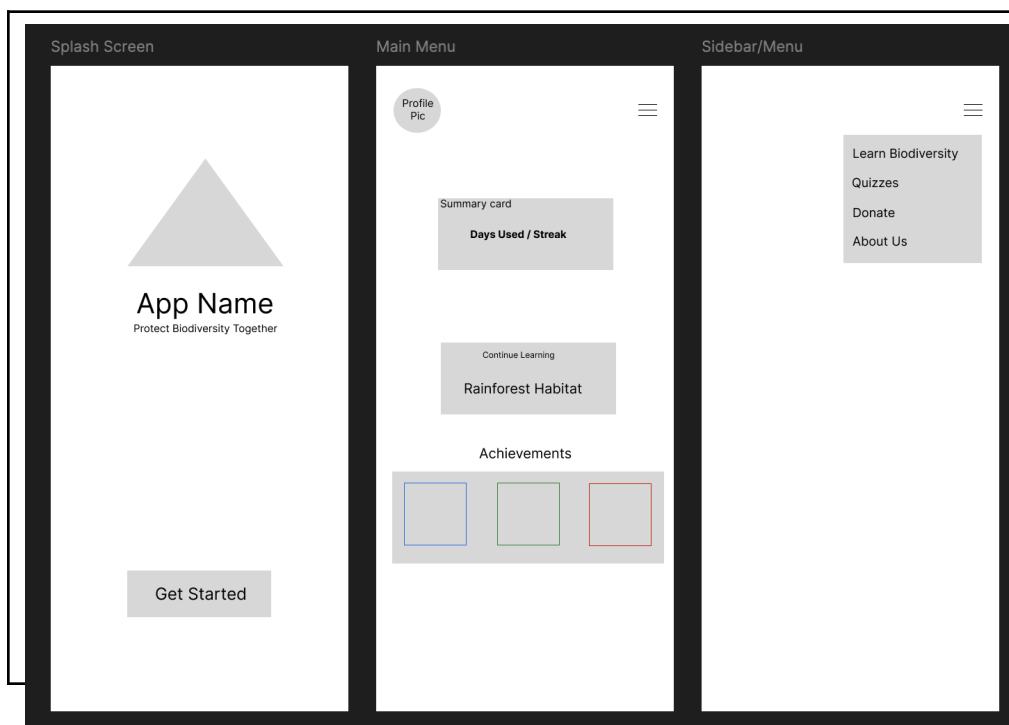
3. The interface will be kept to a minimalist style of design to prevent too much eye-catching elements
4. Navigation will be made easier through implementation of navigation bar at the bottom which will be visible through every action the user takes.
5. The donation function will provide users with multiple methods to make donations, making the function fuss free.
6. In order for users to develop an emotional attachment to the app, users are able to virtually adopt an animal, where they will also be able to see their location in the app.

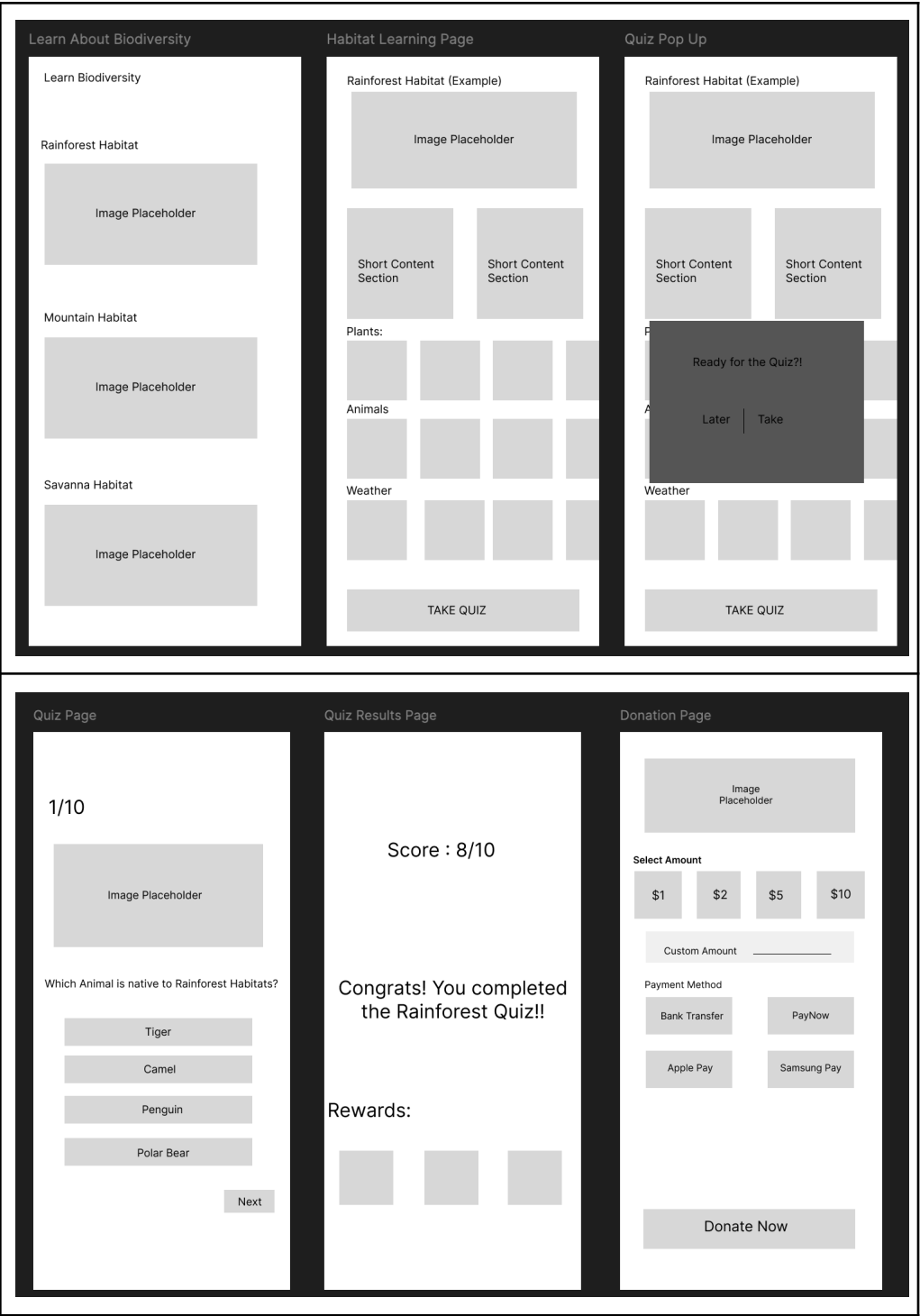
Justification for Design

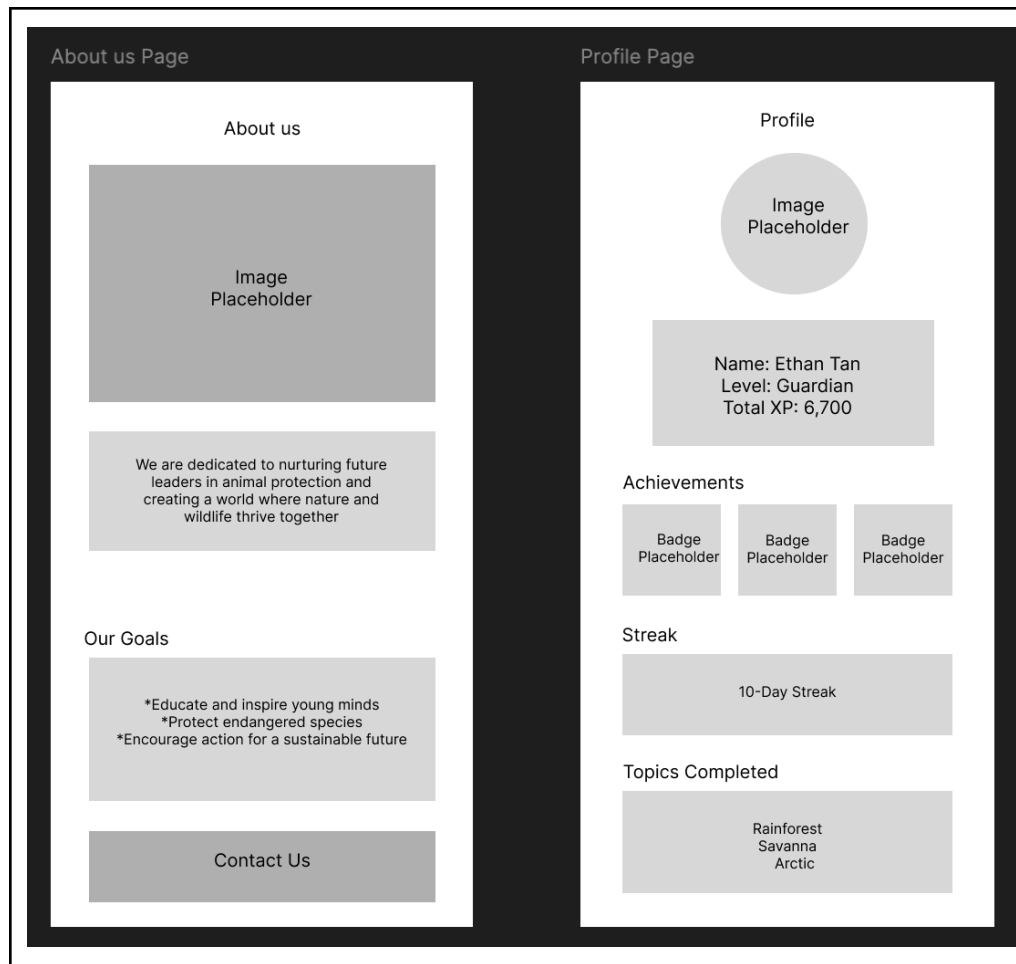
As our SDG is Life on Land, which focuses on biodiversity, the team has decided on using different shades of green and white for the interface. With our primary stakeholders who are students in mind, we will be making the interface look clean and simplified. This design choice will make the app easier to use, making it more attractive as well. Our team has decided to adopt some inspirations from Apple IOS design, such as the soft shapes typically used in IOS devices. By using the color scheme of different shades of green and white, it will connect the app closer to its goal, which is teaching users about biodiversity.

Lo-Fi Wireframes

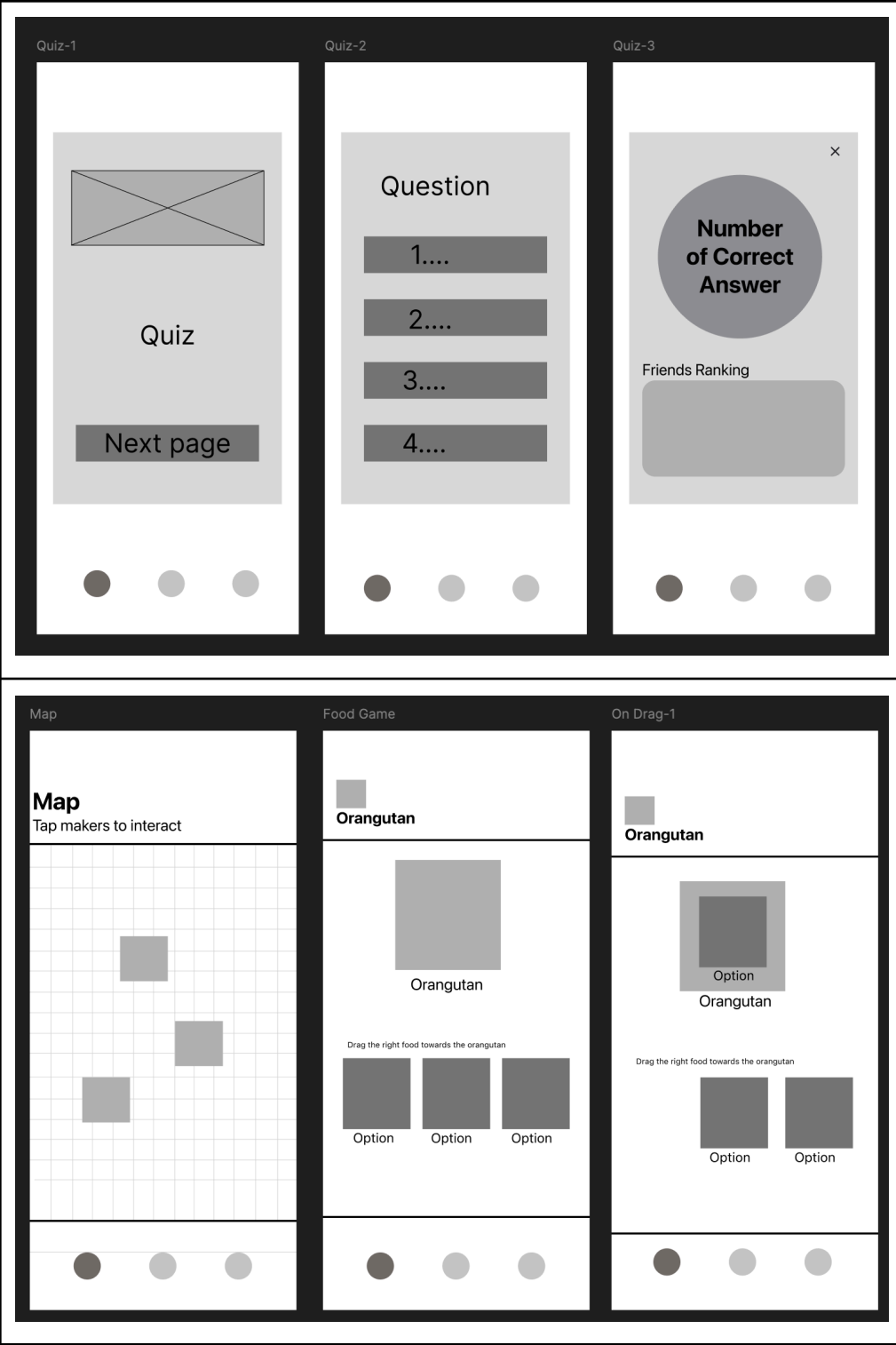
Wireframe design 1

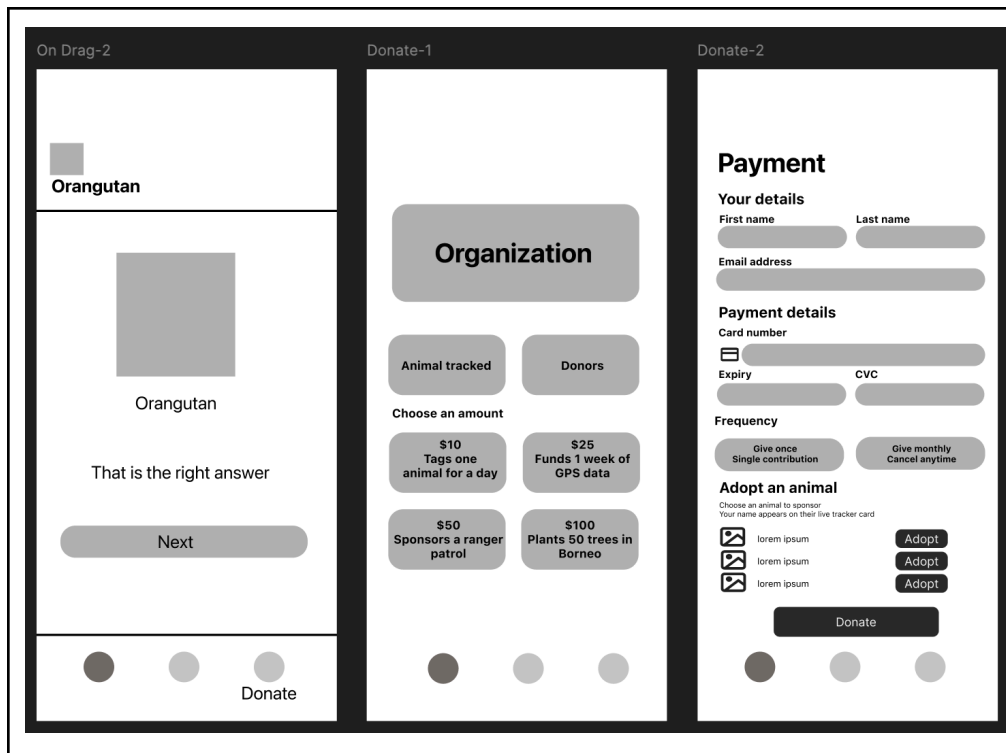




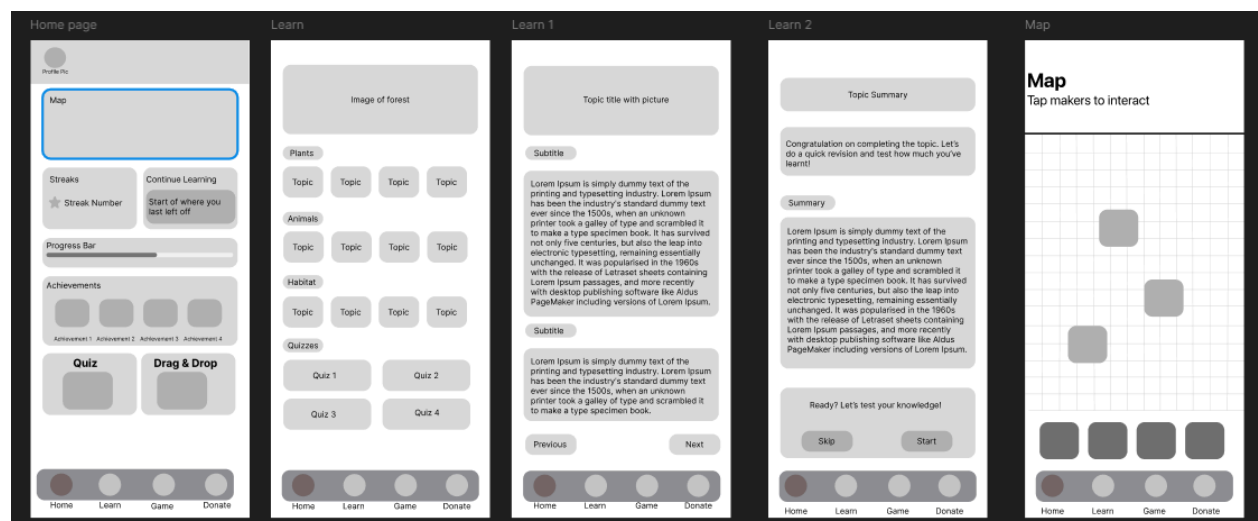


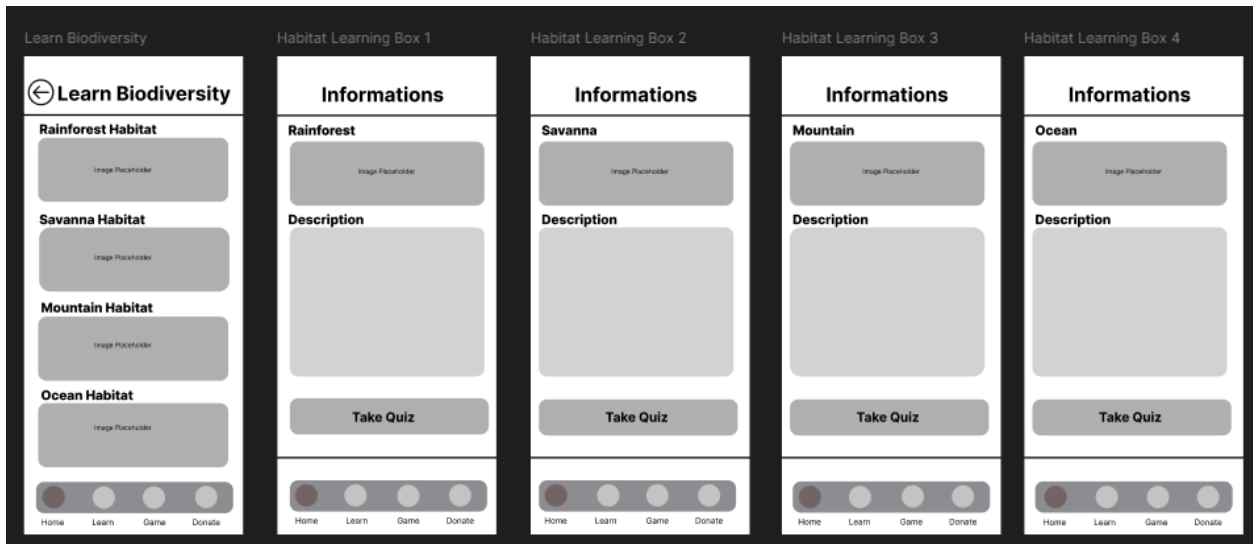
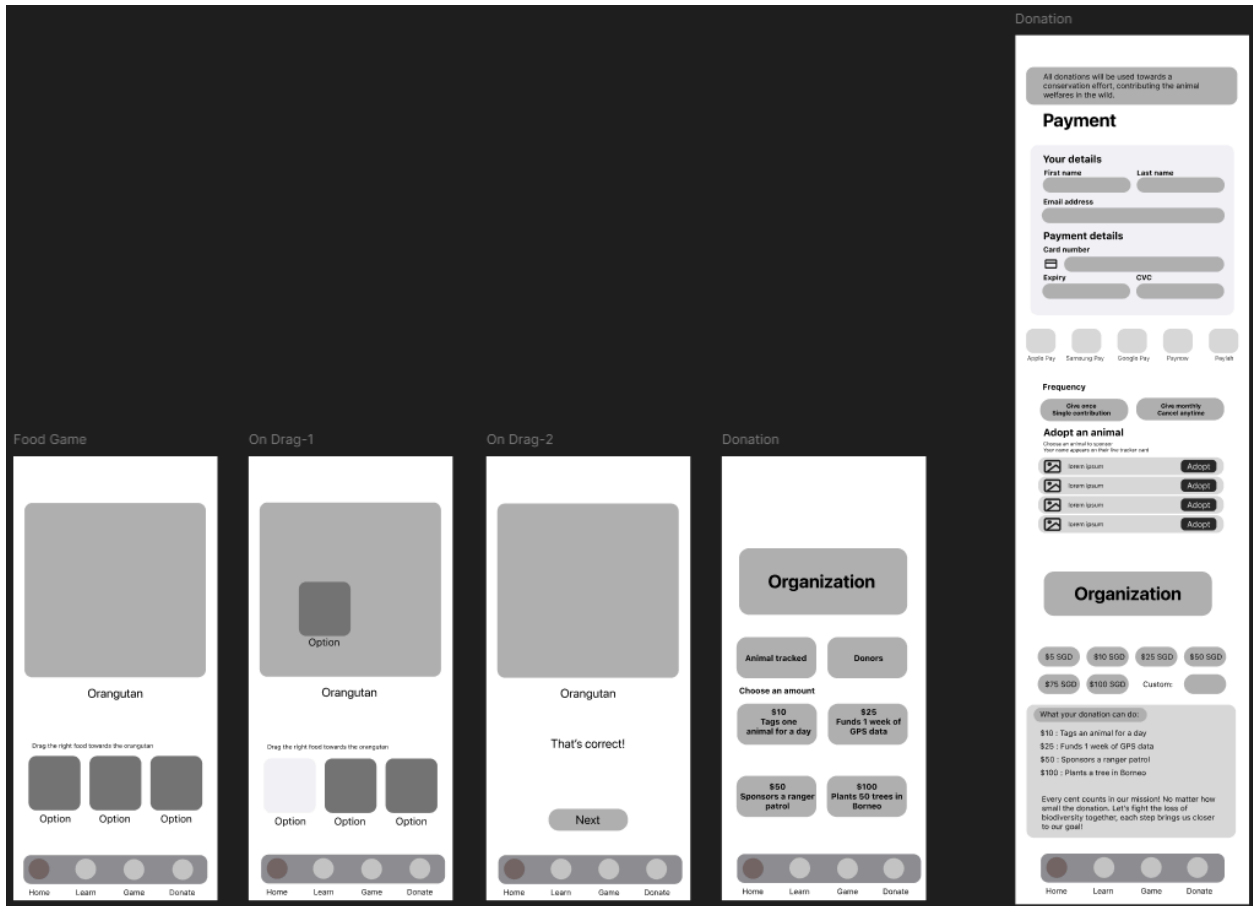
Wireframe design 2

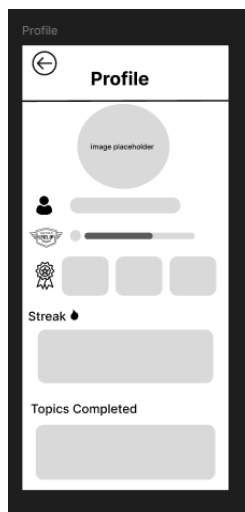
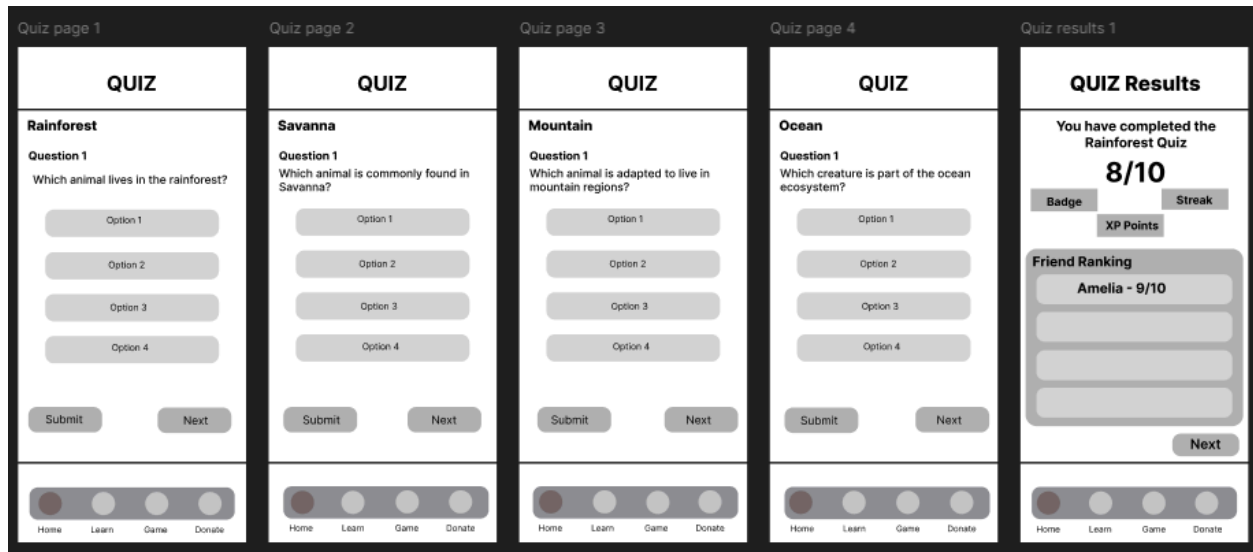




Lo-Fi Design







Interface Development Iterations

In the initial stage of planning, the team came up with a myriad of ideas, ranging from interactive games to integrated videos. We decided to add all of those functions in our application. However, after reviewing the purpose and goal of our project, these functions would appear as being overboard. This thought came along after thinking back that our application is meant to educate our primary stakeholders who are students from Secondary and Tertiary education. As our main goal is to provide our stakeholders with a meaningful way of spending their free time, learning about real-world knowledge which are close to them. Thus, we decided to keep the app clean and easy, covering enough knowledge points without causing users to be addicted to the app in a bad way.

After we laid out what functions the app should have, we started with our wireframe. With our group splitting into 2 teams, 2 distinct designs came to light. The first design was more function oriented, with various options available on each page. The second design was more visual oriented, focusing heavily on background visuals to attract users. After heavy brainstorming, we decided to use the first design as base and integrate some elements from the second design. This resulted in a functional interface, with visuals as buttons for the user instead of the typical word oriented options.

Once the lofi prototype had been created, it was clear how we wanted the functions to work, next was to make sure everything works while we refine the overall design. During the process of creating the hifi prototype, the team worked together relentlessly, making sure the interface was perfectly aligned down to pixels. The first draft of hifi looked simple, but not engaging enough. The team researched and brainstormed different designs used by listed companies. This sparked the idea of using colored minimalist icons, improving the overall visuals without complications. It was the perfect idea to keep our interface simple yet engaging.

When the team had completed the second draft of designs, we reviewed the interface design again. Our team members tested the application and used a “would be nice” perspective to find out what functions are missing. This led to us adding in an additional function, allowing users to make donations in the app. This decision will give users a sense of connection to the app, where they are actively contributing to conservation efforts in real life. Thus, resulted in our final product and concluding our discussions.

System Testing

(What methods will be used to test the system)

Unit testing White-box - individual functions tested in isolation	Every core function was tested individually to confirm it produces the correct output for a given input. This is the first line of defence — bugs caught here are the cheapest and fastest to fix before they affect other parts of the system. Test cases Streaks: streak counter increments by 1 on daily activity, resets to 0 on a missed day, and personal best is never overwritten by a lower value Quiz: correct answer detection returns true/false accurately, timer expiry is handled as a wrong answer, and points per question calculate correctly Topic progress: Completion indicator activates when the user reaches the end of a learning page, a progress percentage updates correctly in the profile section.
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	Map distance: GPS coordinate maths between user location and each animal marker returns an accurate kilometre value Donation: payment amount validation rejects zero and negative values, and one-time vs monthly toggle correctly sets the billing type
Integration testing How features connect and pass data to each other	Once individual units passed, features were combined and tested together to verify that the data flowing between them is correct. This is where connection bugs appear — for example, a quiz completing but the streak not updating. Test cases Quiz → streak: completing the daily quiz increments the streak counter on the same session without needing a restart Topic read → progress: finishing a learning page immediately updates the category progress bar and completed topics list on the profile Streak milestone → achievement: hitting 7 consecutive days automatically triggers and displays the Streak Starter badge on the profile without manual action Donation → adoption unlock: a completed donation immediately makes the adoption animal selection available — no manual approval required Adoption → map: after adopting an animal, that animal's marker on the live map switches from anonymous to named — showing name, tag ID, speed, and distance Adoption → badge: completing an adoption fires the First Adoption achievement badge automatically
Functional testing Black-box · does every feature do what it should	Each feature was tested against its requirements without looking at the underlying code — exactly as a user would experience it. This confirms that every screen, button, and flow produces the right outcome for every possible input. Test cases Streaks: 7-day calendar shows correct filled and empty states, personal best updates when broken record is set, and streak resets to zero on a missed day Quiz:

	all 5 questions load in sequence, correct answers turn green with points shown, wrong answers turn red with an explanation, results screen shows accurate score and star rating Map — non-donor: anonymous animal markers appear and move in real time, no name or tag data is displayed, tapping shows only species type and general biome Map — adopter: adopted animal shows name, tag ID, current speed, direction, and live distance from user, all updating in real time Achievements: all 6 badges display correctly, locked badges show remaining steps, earned badges show the correct unlock date Donation: amount selection, one-time vs monthly toggle, payment form validation, confirmation screen, and adoption selection all function end to end Topic progress: all four category bars (Animals, Plants, Habitat, Quiz) reflect accurate percentages and the completed topics list shows correct dates
Usability testing Real users completing real tasks	Real participants were asked to complete tasks without guidance to identify where the app is confusing, unclear, or requires too many steps. This reveals experience problems that functional testing cannot — a feature can work perfectly but still feel hard to use. Test tasks Task 1: "Start today's quiz and submit your answers" — checking that new users can find and complete the quiz without guidance Task 2: "Find out how far the nearest animal is from you on the map" — verifying the distance display is immediately readable and the map is intuitive to navigate Task 3: "Donate and adopt an animal" — measuring whether the donation-to-adoption flow is clear and users understand what they unlock Task 4: "Read a topic and find your progress on the profile" — checking that users understand the topic progress feature and can navigate between learning and profile pages Task 5: "Find your achievements and see what badge comes next" — testing

	whether locked badge descriptions give users a clear goal
Performance testing Speed, load time, and real-time update reliability	The app was tested under normal and peak load conditions to ensure all screens load quickly and the live map updates reliably. The map is the most demanding feature — it polls satellite GPS data every 15 minutes and must update animal marker positions without freezing the interface. Test cases Screen load time: All major screens (Homepage, Learn, Quiz, Profile, and Donation) must load within 2 seconds when navigated from the bottom menu. Real-time marker refresh: animal position updates every 15 minutes without freezing the map or blocking user interaction Quiz timer accuracy: Countdown must remain consistent and visible throughout all quiz pages, tested while switching between questions Profile load time: streak calendar, progress bars, achievements, and topic list must all render within 1.5 seconds Donation page: payment form must be interactive within 1 second of opening — no delay on amount selection buttons
Compatibility testing iOS, Android, and different screen sizes	The full app was tested across the latest iPhone and Samsung flagship devices on the current iOS 26 and Android 16 operating systems to confirm a consistent experience for all users. The live map and location-based distance display are particularly sensitive to device differences — GPS permission handling varies between iOS and Android. iOS — iPhone 17e, iPhone 17 / Air, iPhone 17 Pro Max All six features function correctly end-to-end across all three iPhone models Location permission prompt appears correctly on first quiz open under iOS 26's updated precise vs approximate location UI iOS 26 Dynamic Island does not overlap the streak calendar header or quiz progress bar on iPhone 17 and Air Proximity alert notification banner displays correctly in iOS 26's Liquid Glass notification design Donation payment form autofill works correctly with iOS 26 Wallet integration on all three models Android — Galaxy S25, Galaxy S25 Ultra, Galaxy S25 Edge Android 16 back gesture navigation does not interrupt the quiz flow mid-question or close the donation form accidentally Proximity alert notification banner displays correctly using Android 16's Live

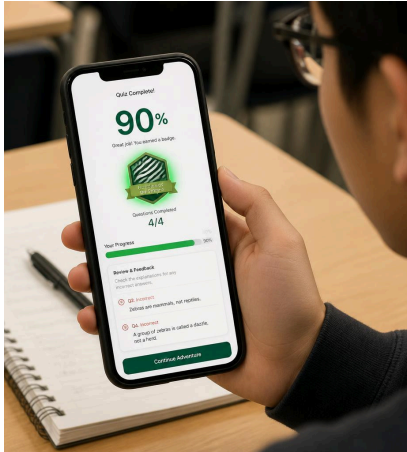
	Updates notification style in the notification shade One UI 8 keyboard does not obscure the donation payment form fields when the keyboard opens on any Galaxy S25 model Galaxy S25 Ultra S Pen stylus input works correctly on animal map markers alongside standard finger touch Galaxy S25 Edge narrow aspect ratio caused the streak 7-day calendar row to overflow — fixed with a fluid flex layout before release
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Storyboard

S/N	Picture	Scenario	Prompt
1		Ethan sits in a lecture room during a lesson on wildlife conservation. While the lecturer explains the importance of protecting endangered species. Ethan opens the Wild Conservation Adventure app on his phone. He's curious about how the app connects to what he's learning, so he started to explore its features. The classroom discussion jot his interest, and he decides to begin his first adventure in the app	A medium close-up shot of a young Chinese Asian male student (early 20s) wearing spectacles, sitting in a lecture room, viewed over his right shoulder at a slight 2 o'clock angle. He holds a smartphone in his right hand, the phone screen is clearly visible and shows the app splash screen with a friendly logo, the tagline "Conservation Adventure", and a large green Start Adventure button. The classroom behind him is softly blurred, rows of desks, a lecturer at the front pointing to a projector screen with an image of a forest and a tiger silhouette, and other students paying attention. The image should be medium close, at a 2 o'clock angle (slightly behind and to the right). Focus on the phone screen and the student's hand, the background is blurred. Clean UI on phone with large tap targets and green accent colour.

2		Ethan is seated at his classroom desk, scrolling through the Wild:Conservation Adventure App during a short break. The mission dashboard presents several regions, each with unique challenges tied to endangered species. His eyes are drawn to the Sumatra card. He started to play the game, as the duration is short and manageable. It felt like a perfect quick challenge before class resumes. With a focused look, he taps to begin his first conservation mission.	A top-down view of a young Chinese Asian male student's hand holding a smartphone on a classroom desk. His index finger hovers over the screen, about to tap a mission card labelled "Sumatra (3 min)" with a small green Recommended badge. The phone screen shows a clean, modern app interface titled "Choose a Mission", displaying a grid of region cards like Sumatra, Borneo, and Java, each with thumbnail maps and small animal icons (tiger, orangutan, and rhino). The background includes a blurred notebook and pen on a wooden desk under warm, natural lighting. Focus on the phone screen and finger interaction: photorealistic style, neutral tones, and shallow depth of field.
3		The level challenges him to restore broken forest patches and reconnect tiger corridors using drag-and-drop mechanics. As he moves a tree icon towards an empty patch of land, the forest begins to regrow, and prey animals return to the area. This kept him engaged, showing how each small action contributes to ecosystem recovery. The timer ticks down from three minutes, but Ethan remains focused, determined to complete the mission before the timer runs out.	A close-up view of a smartphone held by a young Chinese Asian male student wearing spectacles and seated at a classroom desk. The phone screen shows an interactive conservation game level where the player restores forest patches and reconnects tiger corridors using drag-and-drop mechanics. The interface displays a lush green forest map with small icons of trees, rivers, and tiger silhouettes. The student's finger is mid-drag, moving a tree icon toward an empty patch of land. It should show prey animals returning as trees regrow. The UI includes a timer at the top (3 min), a hint button, and an XP bar at the bottom.

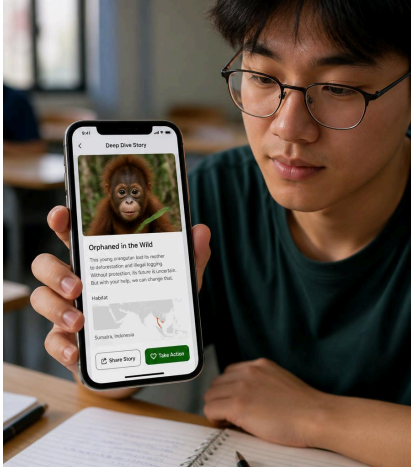
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Ethan sits at his classroom desk, checking the results of his latest conservation quiz on the Wild:Conservation Adventure app. The screen displayed a 90% score and a glowing badge titled "Guardian of the Stripes", symbolising his growing mastery of tiger conservation. He scrolls through the brief explanations for his wrong answers, reviewing what he missed and how to improve next time. The award system made learning feel more rewarding than stressful. Ethan taps "Continue Adventure" to play the next mission

A close-up view of a smartphone held by a young Chinese Asian male student, wearing spectacles, seated at a classroom desk. The phone screen shows a quiz completion interface with four multiple-choice questions completed. The final results screen is visible: a large score "90%" displayed at the top, a badge icon labelled "Guardian of the Stripes" glowing in green, and a horizontal progress bar advancing to 90%. Beneath the score, short one-line explanations appear for incorrect answers. The interface is clean and modern, with rounded cards and a white background. At the bottom, a button reads "Continue Adventure."

5		The game has become addictive for Ethan, as he's been playing it for 10 days. The Wild: Conservation Adventure app greets him with a glowing flame icon and the message "10-Day Streak". He scrolls through the calendar grid, seeing each day marked with green ticks, proof of his steady commitment. A reward preview at the bottom promises bonus XP if he keeps his streak alive. It's a small but satisfying reminder that consistency matters. Feeling motivated, Ethan decides he won't miss tomorrow's session.	A close-up view of a smartphone held by a young Chinese Asian male student wearing spectacles at a classroom desk. The phone screen shows the app's dashboard highlighting a 10-day streak. At the top, a glowing flame icon with the text "10-Day Streak" is displayed. Below it, a small calendar grid shows consecutive days checked off with green ticks. A reward preview card appears at the bottom, showing a badge outline and XP points labelled "Keep streak for bonus." The interface is clean and modern, with rounded cards and a white background.
6		Ethan sits with his classmates after school, excitedly sharing the Wild:Conservation Adventure app on his phone. The screen displays a new Team Challenge, "Restore Sumatra Forest – Timed Mission". His friends lean in, intrigued by the leaderboard showing team scores and progress. The friendly competition sparks laughter and motivation as they plan how to climb to top together.	A wide shot of three students sitting together at a classroom table, one of them (a young Chinese Asian male wearing spectacles) holding a smartphone and showing the screen to his classmates (a Malay male and an Indian male). The phone screen displays a clean, modern app interface with a Team Challenge banner at the top, a leaderboard with team names and scores, and a mission card labelled "Restore Sumatra Forest — Timed Challenge". A button at the bottom reads "Invite Friends". The classmates lean in with interested, excited expressions.

7		The screen shows a photo of an orphaned baby orangutan, its eyes wide and innocent, paired with a short narrative about habitat loss. He's absorbed in the story, imagining the orangutan's struggle to survive. The map icon below the text helps him visualise where the tragedy unfolded, making the issue feel real and close. His expression softens, empathy replaces curiosity. When the "Share Story" and "Take Action" buttons appeared, Ethan felt like he was compelled to do something.	A medium close shot of a smartphone held by a young Chinese Asian male student wearing spectacles at a classroom desk. The phone screen shows a deep-dive story interface with a large photo of an orphaned orangutan at the top, beneath it a short narrative text block (3–4 lines of placeholder text), and a small habitat map icon. At the bottom of the screen are two clear buttons: "Share Story" and "Take Action." The UI is clean, modern, with rounded cards and a white background. The student's face is softly lit, showing a thoughtful, empathetic expression as he reads.
8		Ethan opens the Action Panel, where three clear options appear – Sign Petition, Donate, and Download Classroom Pack. Each button represent, simple and solid way to make a difference. After tapping Donate, a confirmation toast pops up "Thanks – 5 trees funded." A small smile spreads across his face as he realises his actions have real impact beyond the screen. Ethan feels empowered that his learning has transformed into meaningful environmental action.	A close-up view of a smartphone held by a young Chinese Asian male student wearing spectacles at a classroom desk. The phone screen shows the app's Action Panel interface. At the top, three large one-tap buttons are visible: Sign Petition, Donate, and Download Classroom Pack. Below the buttons, a confirmation toast appears with the text: "Thanks — 5 trees funded." The interface is clean and modern, with rounded cards, green accent colors, and a white background

APA Reference List

Leonardo AI had been used in Storyboard to generate pictures.

Contribution Chart

Name	UOW ID	Percentage	Contributed Portion
Sarthak Sinha	9785565	16.67	- Lo-Fi Wireframe 1 - Interface Development Iteration - Overall Design Consideration
Cheung Xi Min	1441164	22.22	- Lo-Fi Wireframe 2 - Lo-Fi Prototype - Hi-Fi Prototype - System Testing
Wang Wenjie	1441607	27.77	- Lo-Fi Prototype - Hi-Fi Prototype - Interface Development Iteration - Overall Design Consideration - Justification for Design
P Reshmah	1349077	22.22	- Storyboard - Lo-Fi Prototype - Hi-Fi Prototype - System testing
Loh Jun Heng	1366956	5.56	- Hi-Fi Prototype
Remo Jhaison Jayakumar	9088477	5.56	- Wireframe 1